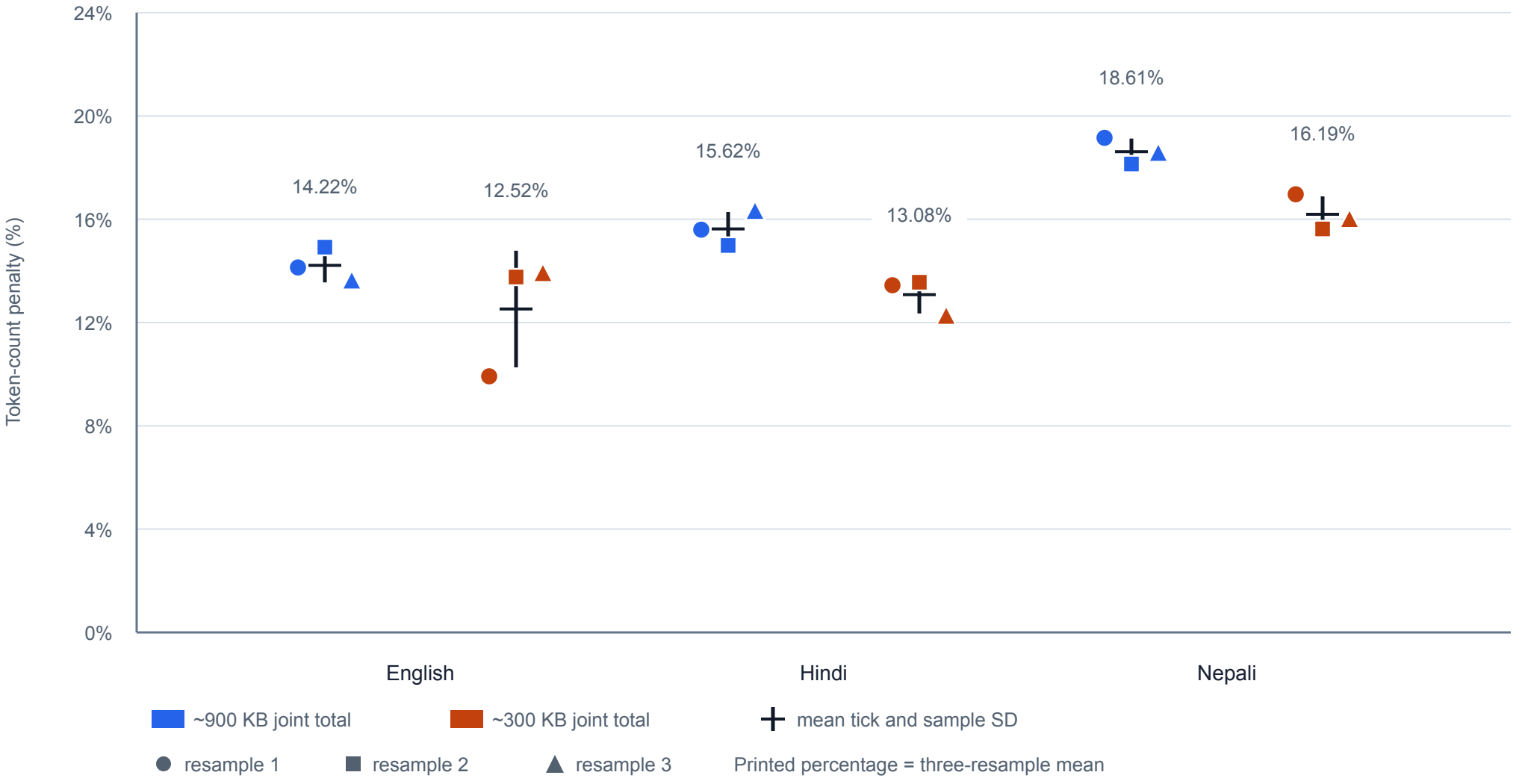
Shared-vocabulary penalty under two total training budgets

Vocabulary 1024; same fixed held-out tests. Color is the joint-training budget; shape is the resample.

Penalty = (shared tokens − dedicated tokens) / dedicated tokens × 100; positive means more tokens.



Replicated summary only. The single-run vocabulary sweep is a separate figure and is not plotted here.

Resamples overlap, so sample SDs describe resample spread only. They are not confidence intervals or significance tests.